

STP 470/Fall 2025
HW (pen-and-paper) *KMeans and PCA*

1. **(50 pts)** Let $\mathbf{X}$ be the (scaled and centered $\Rightarrow$ all its columns have mean 0) $n \times p$ data matrix with $p < n$. Assume that we have the decomposition

$$\mathbf{X}^T \mathbf{X} = \mathbf{U}^T \mathbf{D} \mathbf{U},$$

where $\mathbf{U}$ is an orthogonal matrix, i.e., $\mathbf{U}^T \mathbf{U} = \mathbf{I}$ and $\mathbf{D}$ is a diagonal matrix,

$$\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_p) \text{ with } \lambda_1 > \dots > \lambda_p \geq 0.$$

Show that the sample variance of the first principal component is equal to

$$\frac{\lambda_1}{n-1}.$$

Note: for an $n \times 1$ vector z with zero mean, the sample variance is given by

$$\text{VAR}(z) = \frac{1}{n-1} z^T z.$$

2. **(50 pts.)** Problem 1/ Section 12.6, ISLP, page 552.